

Q₁) $a_1 = 7, a_{n+1} = 7a_n - 3$

$n=1$ case

$$\text{R.H.S} = \frac{1}{2} (13 \times 7^{1-1} + 1) = \frac{1}{2} (14) = 7 = a_1 = \text{L.H.S}$$

$n=1$ case is true 2 cases (S)

$n=p$ (p.e.z) case is true. 2 cases (S)

$$\therefore a_p = \frac{1}{2} (13 \cdot 7^{p-1} + 1) \quad (*)$$

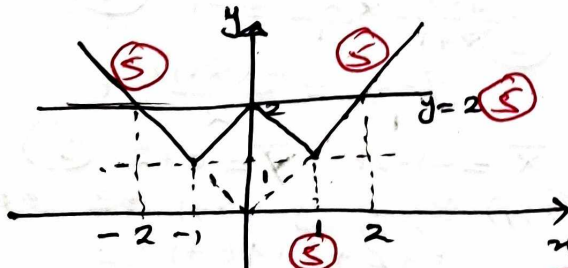
$n=p+1$ case

$$\begin{aligned} \text{L.H.S} = a_{p+1} &= 7a_p - 3 \\ &= 7 \left[\frac{1}{2} (13 \cdot 7^{p-1} + 1) \right] - 3 \\ &= \frac{13 \times 7^p + 7}{2} - 3 \\ &= \frac{1}{2} (13 \times 7^p + 1) \\ &= \text{R.H.S.} \end{aligned}$$

$\therefore n=p+1$ case is true 2 cases

$\therefore$ statement is true for all natural numbers. (S) (25)

Q₂) $y = |1-x| + 1, y = 2$



$$|1-x| \leq 1$$

$$|1-x| + 1 \leq 2$$

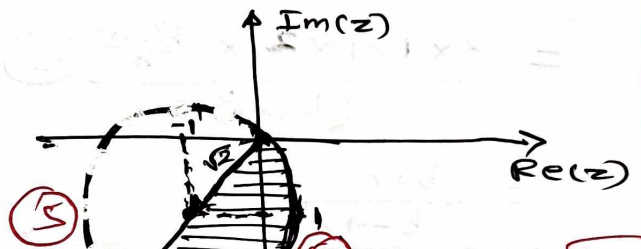
$$\{x\} - 2 \leq x \leq 2 \quad (S)$$

Q₃) $|\bar{z} - 2 + i| \leq \sqrt{2}, -3\pi/4 \leq \text{Arg}(z) \leq 0$

$$|\bar{z} + 1 + i| \leq \sqrt{2}$$

$$|\bar{z} + 1 + i| \leq \sqrt{2} \quad (S)$$

$$|z - (-1-i)| \leq \sqrt{2}$$



$$\begin{aligned} \text{Area of region} &= \frac{1}{2} \times \pi (\sqrt{2})^2 \\ &= \pi \end{aligned}$$

$$\begin{aligned} Q_4) (x^2 + \frac{3k}{\pi})^8 &= \sum_{r=0}^8 {}^8C_r (x^2)^{8-r} \left(\frac{3k}{\pi}\right)^r \\ &= \sum_{r=0}^8 {}^8C_r (3k)^r \pi^{16-3r} \end{aligned}$$

$$x^{-2} \text{ term} \quad -2 = 16 - 3r$$

$$3r = 18 \quad r = 6 \quad (S)$$

$${}^8C_6 (3k)^6$$

$$x^4 \text{ term} \quad 4 = 16 - 3r$$

$$3r = 12 \quad r = 4 \quad (S)$$

$${}^8C_4 (3k)^4$$

$$\begin{aligned} {}^8C_6 (3k)^6 &= {}^8C_4 (3k)^4 \\ (3k)^2 &= \frac{{}^8C_4 (4! 4!)}{{}^8C_6 (6! 2!)} = \frac{6! \times 2!}{4! \times 4!} \end{aligned}$$

$$9k^2 = \frac{5 \times 6}{3 \times 4} = \frac{5}{2}$$

$$k^2 = \frac{5}{18}$$

$$k = \pm \sqrt{\frac{5}{18}} \quad (S)$$

$$\begin{aligned} 5) \lim_{n \rightarrow 0} \frac{(\cos an - \cos bn)(n+a)}{\sin^2(b-a)n} \\ = \lim_{n \rightarrow 0} \frac{2 \sin(\frac{a+b}{2}n) \sin(\frac{b-a}{2}n)(n+a)}{\sin^2(b-a)n} \end{aligned}$$

$$\begin{aligned} &= \frac{2 \lim_{n \rightarrow 0} \frac{\sin(\frac{a+b}{2}n)}{(\frac{a+b}{2}n) \times \lim_{n \rightarrow 0} \frac{\sin(\frac{b-a}{2}n)}{(\frac{b-a}{2}n)} \times \lim_{n \rightarrow 0} (n+a) \times \frac{b^2-a^2}{4}}{\lim_{n \rightarrow 0} \frac{\sin^2(b-a)n}{(b-a)^2 n^2} \times (b-a)^2} \\ &= \frac{2 \times 1 \times 1 \times 2 \times \frac{b^2-a^2}{4}}{(b-a)^2} \\ &= \frac{b+a}{b-a} \end{aligned}$$

$$\begin{aligned} 6) \int_{-\pi/4}^{\pi/4} |\tan x| dx &= \int_{-\pi/4}^0 -\tan x dx + \int_0^{\pi/4} \tan x dx \\ &= + \left[-\ln(\cos x) \right]_{-\pi/4}^0 + \left[\ln(\cos x) \right]_0^{\pi/4} \\ &= \ln \left| \frac{1}{\sqrt{2}} \right| - \ln \left| \frac{\sqrt{2}}{1} \right| \\ &= \ln \sqrt{2} + \ln \sqrt{2} \\ &= 2 \ln \sqrt{2} \end{aligned}$$

$$\begin{aligned} 7) y &= u^2 + 1 \quad u = \sin 2n \\ \frac{dy}{dn} &= 2u \quad \frac{du}{dn} = 2 \cos 2n \\ \frac{dy}{dn} &= \frac{dy}{du} \times \frac{du}{dn} \\ &= 2u \times 2 \cos 2n \end{aligned}$$

$$\begin{aligned} \frac{dy}{dx} &= 4 \sin 2n \cos 2n \\ &= 2 \sin 4n \end{aligned}$$

$$\begin{aligned} x &= \pi/8 \text{ when } n=0 \\ \frac{dy}{dn} \Big|_{n=\pi/8} &= 2 \sin \pi/2 = 2 \end{aligned}$$

$$\begin{aligned} u &= \sin \pi/4 = \frac{1}{\sqrt{2}} \\ \therefore y &= \frac{1}{2} + 1 = \frac{3}{2} \end{aligned}$$

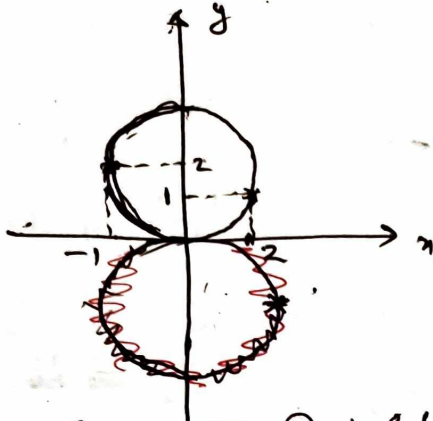
$$\begin{aligned} (x, y) &= (\pi/4, 3/2) \\ (y - 3/2) &= 2(x - \pi/4) \\ 2y - 3 &= 4x - \pi \\ 2y - 4x + (\pi - 3) &= 0 \end{aligned}$$

$$\begin{aligned} &(-1, 2) \\ &N(x, y) \\ &\frac{y-2}{x+1} \times \frac{2}{1} = -1 \\ \frac{y-2}{x+1} &= -\frac{(x+1)}{2} = t \end{aligned}$$

$$\begin{aligned} x &= -1-2t, \quad y = t+2 \\ (x, y) &= (-1-2t, t+2) \\ t+2 + 2+4t+1 &= 0 \\ t &= -1 \end{aligned}$$

$$\begin{aligned} P.N &\equiv (-1+2, -1+2) \\ &\equiv (1, 1) \end{aligned}$$

09)



center of circle is $(0, 1)$

radius of circle is $r = 1$

$$\therefore x^2 + y^2 + 2fy = 0$$

(2,1) lies on the circle.

$$4 + 1 + 2f = 0 \Rightarrow f = -\frac{5}{2}$$

$$x^2 + y^2 - 5y = 0$$

origin $\rightarrow 2x + 1y - \frac{5}{2}(y+1) = 0$

$$4x + 2y - 5y - 5 = 0$$

$$4x - 3y - 5 = 0$$

[25]

10) $\sin(\sin^{-1}(1/5) + \cos^{-1}x) = 1$

$$\sin^{-1}(1/5) + \cos^{-1}x = \sin^{-1}(1)$$

$$= \pi/2$$

$$\sin^{-1}(1/5) = \alpha, \cos^{-1}x = \beta$$

$$\sin \alpha = 1/5, \cos \beta = x$$

$$\alpha + \beta = \pi/2$$

$$\therefore \alpha = \pi/2 - \beta$$

$$\sin \alpha = \sin(\pi/2 - \beta) = \cos \beta$$

$$1/5 = x$$

$$\therefore x = 1/5$$

[25]

011)

$$f(x) = 8x^2 - 6x + k$$

$$x^2 \text{ coefficient} = 8 > 0$$

$$\Delta \leq 0$$

$$36 - 4 \cdot 8 \cdot k \leq 0$$

$$\frac{36}{32} < k$$

$$k > \frac{9}{8}$$

[15]

$$g(x) = 2x^2 + x(2k-1) - k$$

$$\Delta = (2k-1)^2 - 4 \cdot 2 \cdot (-k)$$

$$= 4k^2 - 4k + 1 + 8k$$

$$= 4k^2 + 4k + 1$$

$$= 4(k^2 + k + 1/4)$$

$$= 4(k + 1/2)^2$$

$$= (2k+1)^2 \geq 0$$

$$\Delta \geq 0 \therefore \text{always true.}$$

[20]

$f(x) = 0, g(x) = 0$ solve for x
 common root of both equations

$$8x^2 - 6x + k = 0 \quad (1)$$

$$2x^2 + (2k-1)x - k = 0 \quad (2)$$

$$(1) \cdot (2) \Rightarrow -4(2k-1)x + 4k - 6x + k = 0$$

$$(-8k + 4 - 6)x = -5k$$

$$x = \frac{5k}{8k+2}$$

[15]

$$(1) \Rightarrow 8 \frac{25k^2}{(8k+2)^2} - \frac{30k}{8k+2} + k = 0$$

$$200k^2 - 30k(8k+2) + k(8k+2)^2 = 0$$

$$200k^2 - 240k^2 - 60k + k(64k^2 + 32k + 4) = 0$$

$$-8k^2 + 64k^3 - 56k = 0$$

$$8k(8k^2 - k - 7) = 0$$

$$k = 0 \text{ or } (8k+7)(k-1) = 0$$

$$k = 0 \text{ or } k = -7/8 \text{ or } k = 1$$

$$k > 0 \therefore k = 1$$

$$8x^2 - 6x + 1 = 0$$

$$(4x-1)(2x-1) = 0$$

$$x = 1/4 \text{ or } x = 1/2$$

[5]

$$2x^2 + x - 1 = 0$$

$$(2x - 1)(x + 1) = 0$$

$$x = \frac{1}{2} \text{ or } x = -1 \quad (5)$$

∴ ~~dividing~~

$$x^2 - (-\frac{3}{4})x - \frac{1}{4} = 0$$

$$4x^2 + 3x - 1 = 0 \quad (5)$$

(b) ~~dividing~~

$$f(x) = (x-a)\phi(x) + R \quad (5)$$

where R

$$(x-a) \text{ divides } f(x) \text{ then } R = 0 \quad (5)$$

$$f(x) = (x-a)\phi(x)$$

$x = a$ value

$$f(a) = (a-a)\phi(a) \quad (5)$$

$$f(a) = 0 \quad (15)$$

$$f(x) = x^5 - 3x^4 + ax^3 - 2x^2 + 3x + b$$

$$(x-1) \text{ divides } \therefore f(1) = 0 \quad (5)$$

$$1 - 3 + a - 2 + 3 + b = 0$$

$$a + b = 1 \quad (1) \quad (5)$$

$$(x+1) \text{ divides } \therefore f(-1) = 0 \quad (5)$$

$$-1 - 3 - a - 2 - 3 + b = 0$$

$$-a + b = 9 \quad (2) \quad (5)$$

$$\textcircled{1} + \textcircled{2} \quad 2b = 10 \Rightarrow b = 5$$

$$\therefore a = -4 \quad (5) \quad (25)$$

$$f(x) = x^5 - 3x^4 - 4x^3 - 2x^2 + 3x + 5$$

$$\begin{array}{r} x^3 + 3x^2 - 5x + 1 \\ x^5 - 3x^4 - 4x^3 - 2x^2 + 3x + 5 \\ \hline x^5 + x^3 \\ \hline -3x^4 - 5x^3 \\ -3x^4 - 3x^2 \\ \hline -5x^3 + x^2 \\ -5x^3 - 5x \\ \hline x^2 + 8x \\ x^2 + 1 \\ \hline 8x + 4 \end{array} \quad (5)$$

$$f(x) = 8x + 4$$

$$f(x) = 8x + 4 \div x^2 + 1 \text{ gives } 4 \quad (5) \quad (20)$$

$$\begin{array}{l} \textcircled{1} - 2 \\ \textcircled{2} - 2 \\ \textcircled{3} - 3 \\ \hline \textcircled{4} - 1 \\ \textcircled{5} - 1 \\ \hline 9 \end{array}$$

Ques	Ans	Soln
3 combinations	$1 \times 4 = 4$	$4 \times \frac{4!}{3!} = 16 \quad (5)$
2 combinations	$3 \times 2 = 3$	$3 \times \frac{4!}{2! \times 2!} = 18 \quad (10)$
2 combinations	$3 \times 2 = 18$	$18 \times \frac{4!}{2!} = 216 \quad (10)$
1 combination	$5 \times 4 = 5$	$5 \times 4! = 120 \quad (5)$

(ii) 0 probability 3 combinations

Ques	Ans	Soln
3 combinations	1	$1 \left(\frac{4!}{3!} - \frac{3!}{2!} \right) = 3 \quad (10)$
2 combinations	$3 \times 4 = 12$	$12 \left(\frac{4!}{2!} - \frac{3!}{2!} \right) = 204 \quad (10)$
1 combination	$5 \times 3 = 10$	$10(4! - 3!) = 180 \quad (10)$

$$370 + 387 = 757 \quad (5) \quad (35)$$

$$\textcircled{b} \quad n^3 - (n-1)^3 = n^3 - [n^3 - 3n^2 + 3n - 1]$$

$$= 3n^2 - 3n + 1 \quad (5) \quad (05)$$

$$1^2 - 0^2 = 3 \times 1^2 - 3 \times 1 + 1$$

$$2^2 - 1^2 = 3 \times 2^2 - 3 \times 2 + 1 \quad (5)$$

$$\frac{n^2 - (n-1)^2}{n^2 - (n-1)^2} = \frac{3(n-1)^2 - 3(n-1) + 1}{3n^2 - 3n + 1} \quad (5)$$

$$n^2 = 3[1^2 + 2^2 + \dots + n^2] - 3[1 + 2 + \dots + n] + n$$

$$n^2 = 3 \sum_{r=1}^n r^2 - \frac{3n(n+1)}{2} + n \quad (5)$$

$$3 \sum_{r=1}^n r^2 = n^2 + \frac{3n(n+1)}{2} - n$$

$$\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6} \quad (20)$$

$$1 + (1+2) + (1+2+3) + \dots$$

$$U_r = 1 + 2 + 3 + \dots + r = \frac{r(r+1)}{2} \quad (5)$$

Q12) b)

$$u_r = \frac{r(r+1)}{2}$$

$$\sum_{r=1}^n u_r = \sum_{r=1}^n \frac{r(r+1)}{2}$$

$$= \frac{1}{2} \sum_{r=1}^n r^2 + \frac{1}{2} \sum_{r=1}^n r$$

$$= \frac{n(n+1)(2n+1)}{12} + \frac{n(n+1)}{4}$$

$$= \frac{n(n+1)(2n+1+3)}{12}$$

$$= \frac{n(n+1)(n+2)}{6}$$

$$V_r = \frac{1}{1+2+3+\dots+r}$$

$$= \frac{1}{\frac{r(r+1)}{2}} = \frac{2}{r(r+1)}$$

$$= 2 \left[\frac{1}{r} - \frac{1}{r+1} \right] = \frac{2}{r} - \frac{2}{r+1}$$

$$V_r = f(r) - f(r+1) \quad \text{where } f(r) = \frac{2}{r}$$

$$V_1 = f(1) - f(2)$$

$$V_2 = f(2) - f(3)$$

$$V_{n-1} = f(n-1) - f(n)$$

$$V_n = f(n) - f(n+1)$$

$$\sum_{r=1}^n V_r = f(1) - f(n+1)$$

$$= 2 - \frac{2}{n+1}$$

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n V_r = \lim_{n \rightarrow \infty} \left(2 - \frac{2}{n+1} \right)$$

$$= 2$$

$\therefore$ Answer is 2

Q13) $A = \begin{pmatrix} 1 & 9 \\ 3 & -4 \end{pmatrix}$

$$|A| = -4 - 3 \times 9$$

$$A^{-1} \text{ exists as } |A| \neq 0$$

$$-4 - 3 \times 9 \neq 0$$

$$a \neq -4/3$$

$$A^{-1} = \frac{1}{-4-3 \times 9} \begin{pmatrix} -4 & -9 \\ -3 & 1 \end{pmatrix}$$

$$= \frac{1}{4+3 \times 9} \begin{pmatrix} 4 & 9 \\ 3 & -1 \end{pmatrix}$$

$$A^2 = \begin{pmatrix} 1 & 9 \\ 3 & -4 \end{pmatrix} \begin{pmatrix} 1 & 9 \\ 3 & -4 \end{pmatrix} = \begin{pmatrix} 3 \times 1 + 9 \times 3 & -4 \times 1 - 9 \times 4 \\ -9 \times 3 + 3 \times -4 & 3 \times 9 + (-4) \times (-4) \end{pmatrix}$$

$$= \begin{pmatrix} 3 \times 1 + 9 \times 3 & -4 \times 1 - 9 \times 4 \\ -9 \times 3 + 3 \times -4 & 3 \times 9 + (-4) \times (-4) \end{pmatrix} + 3 \begin{pmatrix} 1 & 9 \\ 3 & -4 \end{pmatrix} - 10 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \times 1 + 9 \times 3 & 0 \\ 0 & 3 \times 9 + (-4) \times (-4) \end{pmatrix} = 0$$

$$\therefore 3 \times 9 - 6 = 0$$

$$9 = 2$$

$$A = \begin{pmatrix} 1 & 2 \\ 3 & -4 \end{pmatrix}, \quad A^{-1} = \frac{1}{10} \begin{pmatrix} 4 & 2 \\ 3 & -1 \end{pmatrix}$$

$$A^2 + 3A - 10I = 0$$

$$(A + 5I)(A - 2I) = 0$$

$$\therefore B = A - 2I$$

$$B = \begin{pmatrix} 1 & 2 \\ 3 & -4 \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ 3 & -6 \end{pmatrix}$$

(b) $z = x + iy$
 $\bar{z} = x - iy$

$$z\bar{z} = (x + iy)(x - iy) = x^2 + y^2$$

$$= (x^2 + y^2) = |z|^2$$

$$|z| = \sqrt{x^2 + y^2} \geq x = \operatorname{Re}(z)$$

$$|z+w|^2 = (z+w)(\bar{z}+\bar{w})$$

$$= (z+w)(\bar{z}+\bar{w})$$

$$= |z|^2 + z\bar{w} + w\bar{z} + |w|^2$$

$$= |z|^2 + z\bar{w} + \bar{z}w + |w|^2$$

$$= |z|^2 + 2\operatorname{Re}(z\bar{w}) + |w|^2$$

$$\leq |z|^2 + 2|z||w| + |w|^2$$

$$= |z|^2 + 2|z||w| + |w|^2$$

$$= (|z| + |w|)^2$$

(13) $\therefore |z+w| \leq |z|+|w|$
 $(|z|+|w|)^2 = |z|^2 + 2|z||w| + |w|^2$
 $\leq |z|^2 + (|z|^2 + |w|^2) + |w|^2$
 $[(|z|-|w|)^2 \geq 0]$
 $|z|^2 + |w|^2 \geq 2|z||w|$
 $(|z|+|w|)^2 \leq 2|z|^2 + 2|w|^2$ (5) [10]

$|z_0+4| + |z_0-4| \geq |z_0+4+z_0-4|$ (5)

$10 \geq |2z_0|$
 $5 \geq |z_0|$ (5) ①

$(|z_0+4| + |z_0-4|)^2 \leq 2|z_0+4|^2 + 2|z_0-4|^2$
 $100 \leq 2[|z_0|^2 + 4z_0 + 4\bar{z}_0 + 16 + |z_0|^2 - 4z_0 - 4\bar{z}_0 + 16]$ (5)

$50 \leq 2[2|z_0|^2 + 32]$ (5)

$18 \leq 2|z_0|^2$

$9 \leq |z_0|^2$

$3 \leq |z_0|$ (5) ②

①, ② $\Rightarrow$

$3 \leq |z_0| \leq 5$ [25]

(F) $z_1 = -1+i = \sqrt{2}(-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i)$
 $= \sqrt{2}(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4})$ (5)
 $z_2 = 1+\sqrt{3}i = 2(\frac{1}{2} + \frac{\sqrt{3}}{2}i)$
 $= 2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$ (5)

$\frac{z_1}{z_2} = \frac{-1+i}{1+\sqrt{3}i} = \frac{(-1+i)(1-\sqrt{3}i)}{(1+\sqrt{3}i)(1-\sqrt{3}i)}$
 $= \frac{(-1+\sqrt{3}) + i(\frac{1+\sqrt{3}}{4})}{4}$ (5)

$\text{Re}(\frac{z_1}{z_2}) = \frac{-1+\sqrt{3}}{4}$ (5) $\text{Im}(\frac{z_1}{z_2}) = \frac{1+\sqrt{3}}{4}$ (5)

$\frac{z_1}{z_2} = \frac{\sqrt{2}(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4})}{2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})}$
 $= \frac{1}{\sqrt{2}}(\cos \frac{5\pi}{12} + i \sin \frac{5\pi}{12})$ (5)

$\therefore \frac{1}{\sqrt{2}} \cos \frac{5\pi}{12} = \frac{-1+\sqrt{3}}{4}$ (5)

$\cos \frac{5\pi}{12} = \frac{-1+\sqrt{3}}{2\sqrt{2}} = \frac{1}{4}(-\sqrt{2}+\sqrt{6})$ (5)

$= \frac{1}{4}(\sqrt{6}-\sqrt{2})$ [35]

Q4) $f(m) = x^2 - \frac{8}{m-1}$ (5)
 $f'(m) = 2x + 8(\frac{1}{(m-1)^2})$ (5)
 $= 2x + \frac{8}{(m-1)^2}$ [10]

$f'(m) = 0 \Rightarrow 2x + \frac{8}{(m-1)^2} = 0$

$2x(m^2-2m+1) = -8$ (5)

$2x^3 - 4x^2 + 2x + 8 = 0$

$x^3 - 2x^2 + x + 4 = 0$

$(x+1)(x^2-3x+4) = 0$

$(x+1)(x^2-3x+4) = 0$

$\therefore x = -1$ (5)

$x = -1$ or $y = 1+4 = 5$

Sign	$x < -1$	$-1 < x < 1$	$1 < x < \infty$
$f'(x)$	(-)	(+)	(+)
Graph			

$\therefore (-1, 5)$ is a local minimum (5)

$(-\infty, -1]$ is increasing (5)

$[1, \infty)$ is increasing (5)

For $x=1$ (5) [40]

For $x=1$ (5)

$\lim_{x \rightarrow \infty} (x^2 - \frac{1}{x-1}) = \lim_{x \rightarrow \infty} \frac{x^3(1 - \frac{1}{x} - \frac{1}{x^2})}{x(\frac{1}{x})}$

$= +\infty$ (5)

Ans $f''(m) = 0 \Rightarrow$

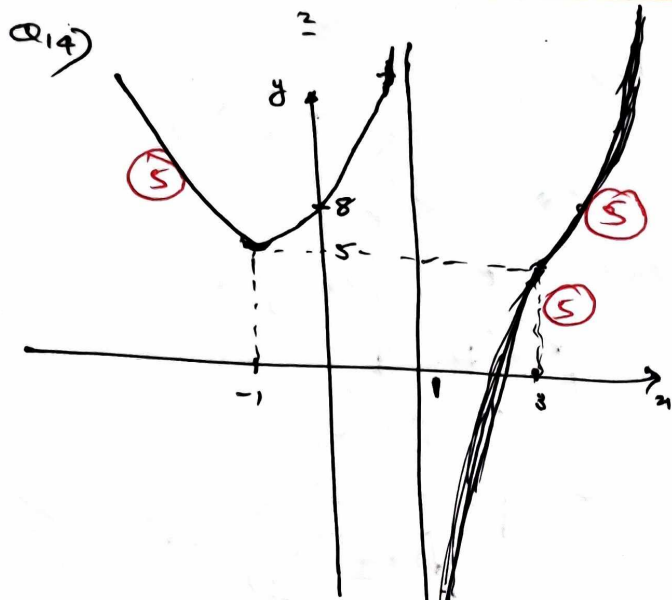
$2 - \frac{16}{(m-1)^3} = 0$

$(m-1)^3 = 8 \Rightarrow m-1 = 2$

$f(3) = 5$ $x-1 = 2 \Rightarrow x = 3$ (5)

Sign	$x < 1$	$1 < x < 3$	$x > 3$
$f'(x)$	(+)	(-)	(+)
Graph			

$(3, 5)$ is a local maximum (5)



(k, ∞) - range of y for $x > k$
 $k=1$ (S) [50]

(b)

$A = \pi \sqrt{4r^2 - x^2}$ (S)

$\frac{dA}{dx} = \pi \left[\frac{1}{2} (4r^2 - x^2)^{-1/2} x - 2x \right]$ (S) [10]

$= \frac{-\pi x}{\sqrt{4r^2 - x^2}} + \sqrt{4r^2 - x^2}$

$\frac{dA}{dx} = \frac{4r^2 - 2x^2}{\sqrt{4r^2 - x^2}}$ (S)

$\frac{dA}{dx} = 0 \Rightarrow 4r^2 - 2x^2 = 0$ (S)
 $x^2 = 2r^2$
 $x = \sqrt{2}r$ (S)

Sign	$0 < x < \sqrt{2}r$	$\sqrt{2}r < x < 2r$
$\frac{dA}{dx}$	(+)	(-)
min	↗ (S)	↘ (S)

$\therefore x = \sqrt{2}r$ is max.

$A_{\max} = \sqrt{2}r \sqrt{4r^2 - 2r^2}$ (S)

$= \sqrt{2}r \cdot \sqrt{2r^2}$

$= \sqrt{2}r \cdot \sqrt{2}r$

$= 2r^2$ (S) [40]

(5)

$2 = A(n-1)(n^2+1) + B(n^2+1) + Cn(n-1)^2$

$[n^3]: 0 = A + C$ (1)

$[n^2]: 0 = -A + B - 2C$ (2) (S)

$[n]: 0 = A + C$ (3)

$[x^0]: 2 = -A + B$ (4)

(2), (4) $\Rightarrow 0 = 2 - 2C \Rightarrow C = 1$ (S)

(3) $\Rightarrow A = -1$ (S)

(4) $\Rightarrow B = 1$ (S)

$\frac{2}{(n-1)^2(n^2+1)} = \frac{A}{n-1} + \frac{B}{(n-1)^2} + \frac{Cn}{n^2+1}$
 $= \frac{-1}{n-1} + \frac{1}{(n-1)^2} + \frac{n}{n^2+1}$ (S)

$\int_2^3 \frac{2}{(n-1)^2(n^2+1)} dn = \int_2^3 \frac{1}{n-1} dn + \int_2^3 \frac{1}{(n-1)^2} dn + \frac{1}{2} \int_2^3 \frac{2n}{n^2+1} dn$ (S)

$= \left[\ln|n-1| \right]_2^3 + \left[\frac{(n-1)^{-1}}{-1} \right]_2^3 + \frac{1}{2} \left[\ln|n^2+1| \right]_2^3$ (S)

$= \ln\left(\frac{1}{2}\right) - \left[\frac{1}{2} - 1\right] + \frac{1}{2} \ln\left(\frac{10}{5}\right)$ (S)

$= -\ln 2 + \frac{1}{2} + \frac{1}{2} \ln 2$ (S)

$= \frac{1}{2} - \frac{1}{2} \ln 2 = a + b \ln 2$

$a = \frac{1}{2}, b = -\frac{1}{2}$ (S) [55]

(b) $\int_0^1 \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx = \int_0^{\frac{\pi}{4}} \sin^{-1}\left(\frac{2 \tan \theta}{1 + \tan^2 \theta}\right) \sec^2 \theta d\theta$ (S)

$\left[x = \tan \theta \right] = \int_0^{\frac{\pi}{4}} 2\theta \sec^2 \theta d\theta$

$\frac{dx}{d\theta} = \sec^2 \theta$ (S)

$x=0 \Rightarrow \theta=0$

$x=1 \Rightarrow \theta=\frac{\pi}{4}$ (S)

$= 2 \left[\theta \tan \theta \right]_0^{\frac{\pi}{4}} + 2 \int_0^{\frac{\pi}{4}} \tan \theta d\theta$ (S)

$= 2 \left[\frac{\pi}{4} \cdot 1 + 2 \ln|1 + \tan \theta| \right]_0^{\frac{\pi}{4}}$ (S)

$= \frac{\pi}{2} + 2 \ln\left(\frac{\sqrt{2}}{1}\right) = \frac{\pi}{2} - \ln 2$ (S) [40]

$$\begin{aligned}
 (c) \int \frac{1}{\sqrt{a^2 + x^2}} dx & \quad x = a \tan \theta \\
 & \quad \frac{dx}{d\theta} = a \sec^2 \theta \\
 & = \int \frac{1}{\sqrt{a^2 + a^2 \tan^2 \theta}} \cdot a \sec^2 \theta d\theta \\
 & = \int \frac{1}{a \sec \theta} \cdot a \sec^2 \theta d\theta \\
 & = \int \sec \theta d\theta \\
 & = \ln |\sec \theta + \tan \theta| + C \\
 & = \ln \left| \sqrt{1 + \frac{x^2}{a^2}} + \frac{x}{a} \right| + C \\
 & = \ln \left| \sqrt{a^2 + x^2} + x \right| - \ln a + C \\
 & = \ln \left| \sqrt{a^2 + x^2} + x \right| + C
 \end{aligned}$$

$$\int \frac{x}{\sqrt{a^2 + x^2}} dx = \sqrt{a^2 + x^2} \cdot x - \int x \cdot \frac{x}{\sqrt{a^2 + x^2}} dx$$

$$I = x \sqrt{a^2 + x^2} - \int \frac{x^2}{\sqrt{a^2 + x^2}} dx$$

$$= x \sqrt{a^2 + x^2} - \int \frac{x^2 + a^2 - a^2}{\sqrt{a^2 + x^2}} dx$$

$$= x \sqrt{a^2 + x^2} - \int \sqrt{a^2 + x^2} dx + \int \frac{a^2}{\sqrt{a^2 + x^2}} dx$$

$$2I = x \sqrt{a^2 + x^2} + a^2 \ln \left| \sqrt{a^2 + x^2} + x \right| + C$$

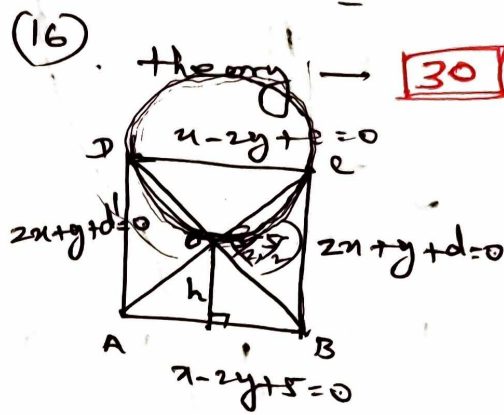
$$I = \frac{1}{2} \left[x \sqrt{a^2 + x^2} + a^2 \ln \left| \sqrt{a^2 + x^2} + x \right| + C \right]$$

$$\int \sqrt{4 + x^2} dx = \frac{1}{2} \left[x \sqrt{4 + x^2} + 4 \ln \left| \sqrt{4 + x^2} + x \right| \right]_0^1$$

$$= \frac{1}{2} \left[\sqrt{5} + 4 \ln(\sqrt{5} + 1) - 0 - 4 \ln 2 \right]$$

$$= \frac{\sqrt{5}}{2} + 2 \ln(\sqrt{5} + 1) - 2 \ln 2$$

$$= \frac{\sqrt{5}}{2} + 2 \ln \left(\frac{\sqrt{5} + 1}{2} \right)$$



$$h = \left| \frac{5/2 - 5 + 5}{\sqrt{1+4}} \right| = \frac{5/2}{\sqrt{5}} = \frac{\sqrt{5}}{2}$$

$$\left| \frac{5/2 - 5 + d}{\sqrt{5}} \right| = \frac{\sqrt{5}}{2}$$

$$\left| \frac{5/2 + d}{\sqrt{5}} \right| = \frac{\sqrt{5}}{2}$$

$$\therefore c = 5 \text{ or } 0$$

$$DC \rightarrow x - 2y = 0$$

$$\left| \frac{5 + 5/2 + d}{\sqrt{5}} \right| = \frac{\sqrt{5}}{2}$$

$$\frac{15}{2} + d = \pm \frac{\sqrt{5}}{2}$$

$$d = -10 \text{ or } -5$$

$$AD \rightarrow 2x + y - 10 = 0$$

$$BC \rightarrow 2x + y - 5 = 0$$

DC is the line passing through D and C.

$$x - 2y = 0 \quad (1) \quad 2x + y - 5 = 0 \quad (2)$$

$$(1) \times 2 \Rightarrow 4x - 4y = 0 \\
 \quad \quad \quad y = 1 \quad \quad \quad C = (2, 1) \\
 \quad \quad \quad x = 2$$

$$x - 2y = 0 \quad (1) \quad 2x + y - 10 = 0 \quad (2)$$

$$(1) \times 2 \Rightarrow 5y = 10 \\
 \quad \quad \quad y = 2 \quad \quad \quad D = (4, 2) \\
 \quad \quad \quad x = 4$$

$\therefore$ area of rectangle

$$(x-2)(x-4) + (y-1)(y-2) = 0$$

$$x^2 + y^2 - 6x - 3y + 10 = 0$$

$$Q16) \quad 2x+y-5=0 \quad \text{--- (1)}$$

$$x-2y+5=0 \quad \text{--- (2)}$$

$$(2) \cdot (1) \Rightarrow 2(2y-5)+y-5=0$$

$$5y-15=0$$

$$y=3$$

$$x=1$$

$$B \equiv (1, 3)$$

Q17) ~~Find the equation of the line passing through the point (1, 3) and perpendicular to the line 2x+y-5=0.~~

$$y-3=m(x-1) \quad \text{--- (1)}$$

$$x^2+y^2-6x-3y+10=0$$

$$\text{centre} \equiv (3, \frac{3}{2}) \quad r = \sqrt{9 + \frac{9}{4} - 10}$$

$$= \sqrt{\frac{45-40}{4}}$$

$$= \frac{\sqrt{5}}{2}$$

$$\frac{|\frac{3}{2}-3-m(3-1)|}{\sqrt{1+m^2}} = \frac{\sqrt{5}}{2}$$

$$\left(-\frac{3}{2}-2m\right)^2 = \frac{5(1+m^2)}{4}$$

$$\frac{9}{4} + 6m + 4m^2 = \frac{5+5m^2}{4}$$

$$9 + 24m + 16m^2 = 5 + 5m^2$$

$$11m^2 + 24m + 4 = 0 \quad \text{--- (2)}$$

$$(11m+2)(m+2)=0$$

$$m = -2 \text{ or } m = -\frac{2}{11} \quad \text{--- (3)}$$

$$\therefore y-3 = -\frac{2}{11}(x-1) \quad \text{--- (4)}$$

$$\text{Q18) BC} \rightarrow (y-3 = -2(x-1)) \quad \text{--- (5)}$$

$$y+2x-5=0 \quad \text{--- (6)}$$

$$S_1 = x^2+y^2-3x-4y+5=0$$

$$g_1 = -\frac{3}{2}, f_1 = -2, c_1 = 5$$

$$S = x^2+y^2-6x-3y+10=0$$

$$g = -3, f = -\frac{3}{2}, c = 10$$

$$2gg_1 + 2ff_1 = 2(-3)(-\frac{3}{2}) + 2(-\frac{3}{2})(-2) \quad \text{--- (7)}$$

$$= 9 + 6 = 15$$

$$c+c_1 = 5+10 = 15 \quad \text{--- (8)}$$

$$\therefore 2gg_1 + 2ff_1 = c+c_1 \quad \text{--- (9)}$$

$$\therefore \text{The circles are orthogonal.} \quad \text{--- (10)}$$

Q17)

$$\sin(A+B) = \sin A \cos B + \cos A \sin B \quad \text{--- (1)}$$

$$B = -B \text{ or } \sin(-B)$$

$$\sin(A-B) = \sin A \cos(-B) + \cos A \sin(-B) \quad \text{--- (2)}$$

$$= \sin A \cos B - \cos A \sin B \quad \text{--- (10)}$$

$$\sin(A+B) \cdot \sin(A-B) = (\sin A \cos B + \cos A \sin B)(\sin A \cos B - \cos A \sin B) \quad \text{--- (3)}$$

$$= \sin^2 A \cos^2 B - \cos^2 A \sin^2 B$$

$$= \sin^2 A (1 - \sin^2 B) - (1 - \sin^2 A) \sin^2 B \quad \text{--- (4)}$$

$$= \sin^2 A - \sin^2 B \quad \text{--- (5)} \quad \text{--- (15)}$$

$$\frac{\sin^2 A - \sin^2 B}{\sin A \cos A - \cos B \sin B} = \frac{2 \sin(A+B) \cdot \sin(A-B)}{2 \sin A \cos A - 2 \cos B \sin B} \quad \text{--- (6)}$$

$$= \frac{2 \sin(A+B) \sin(A-B)}{\sin 2A - \sin 2B} \quad \text{--- (7)}$$

$$= \frac{2 \sin(A+B) \sin(A-B)}{2 \sin(A-B) \cos(A+B)} \quad \text{--- (8)}$$

$$= \tan(A+B) \quad \text{--- (15)}$$

$$(b) \quad T_m = 1 - \sqrt{3} + 2 \sin m (\cos m - \sqrt{3} \sin m) \quad \text{--- (9)}$$

$$= 1 - \sqrt{3} + 4 \sin m (\frac{1}{2} \cos m - \frac{\sqrt{3}}{2} \sin m) \quad \text{--- (10)}$$

$$= 1 - \sqrt{3} + 4 \sin m (\sin \frac{\pi}{6} \cos m - \cos \frac{\pi}{6} \sin m) \quad \text{--- (11)}$$

$$= 1 - \sqrt{3} + 4 \sin m \cos(m - \frac{\pi}{6}) \quad \text{--- (12)}$$

$$= 1 - \sqrt{3} + 2 [\sin(2m - \frac{\pi}{6}) + \sin \frac{\pi}{6}] \quad \text{--- (13)}$$

$$= 1 - \sqrt{3} + 2 [\sin(2m - \frac{\pi}{6}) + \frac{1}{2}] \quad \text{--- (14)}$$

$$= 1 - \sqrt{3} + 2 \sin(2m - \frac{\pi}{6}) + 1 \quad \text{--- (15)}$$

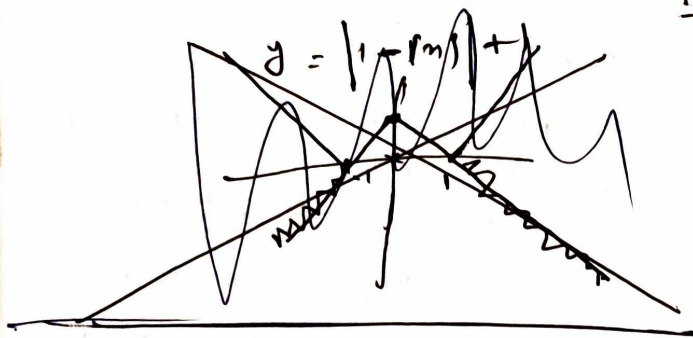
$$= (2 - \sqrt{3}) + 2 \sin(2m - \frac{\pi}{6}) \quad \text{--- (16)}$$

$$= A + B \sin(2m - \alpha)$$

$$\text{Q18) } A = 2 - \sqrt{3} \quad \text{--- (17)}$$

$$B = 2$$

$$\alpha = \frac{\pi}{6} \quad \text{--- (18)}$$



Q17)

(c)

$$\frac{\sin A}{\sin C} = \frac{\sin(A-B)}{\sin(B-C)}$$

$$2 \sin A \sin(B-C) = 2 \sin C \sin(A-B) \quad (5)$$

$$\cos(A+C-B) - \cos(A+B-C) = \cos(C+B-A) - \cos(C+A-B) \quad (5)$$

$$\cos(\pi-2B) - \cos(\pi-2C) = \cos(\pi-2A) - \cos(\pi-2B) \quad (5)$$

$$-\cos 2B + \cos 2C = -\cos 2A + \cos 2B$$

$$\cos 2C + \cos 2A = 2 \cos 2B \quad (5)$$

$$1 - 2 \sin^2 C + 1 - 2 \sin^2 A = 2(1 - 2 \sin^2 B) \quad (20)$$

$$\sin^2 C + \sin^2 A = 2 \sin^2 B \quad (5)$$

$$k^2 c^2 + k^2 a^2 = 2k^2 b^2 \quad (5)$$

$$\left[\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} = k \right]$$

$$c^2 + a^2 = 2b^2 \quad (5)$$

$$c^2 - b^2 = b^2 - a^2$$

$\therefore a^2, b^2, c^2$ are in arithmetic progression. (5)

Q18) Find the range of T .

(20)

Q17)

$$T(x) = (2-\sqrt{3}) + 2 \sin(2x - \pi/6)$$

$$T_{\max} = 2 - \sqrt{3} + 2 = 4 - \sqrt{3} \quad (5)$$

$$\sin(2x - \pi/6) = 1$$

$$2x - \pi/6 = \pi/2$$

$$x = \pi/3 \quad (5)$$

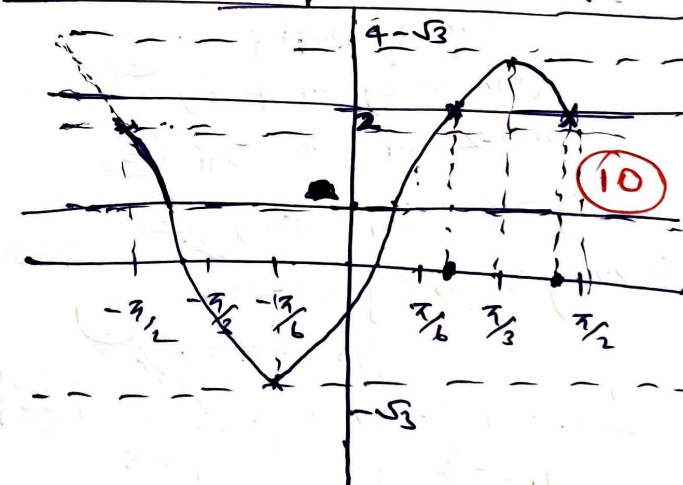
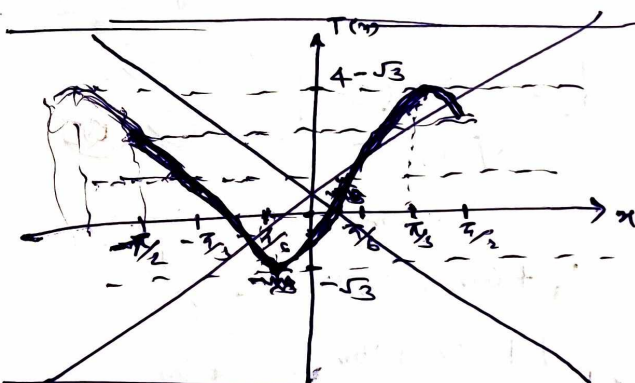
$$T_{\min} = 2 - \sqrt{3} - 2 = -\sqrt{3} \quad (5)$$

$$\sin(2x - \pi/6) = -1$$

$$2x - \pi/6 = -\pi/2$$

$$x = -\pi/6 \quad (5)$$

$$T \text{ - range: } -\sqrt{3} \leq T \leq 4 - \sqrt{3}$$



$$T = 2 \sin(2x)$$

$$2 = 2 - \sqrt{3} + 2 \sin(2x - \pi/6)$$

$$\sin(2x - \pi/6) = \frac{\sqrt{3}}{2} = \sin \pi/3 \quad (5)$$

$$2x - \pi/6 = n\pi + (-1)^n \pi/3$$

$$2x = n\pi + (-1)^n \pi/3 + \pi/6$$

$$x = \frac{n\pi}{2} + (-1)^n \pi/6 + \pi/12$$

$$\therefore x = \pi/6 + \pi/12 = \frac{3\pi}{12} = \pi/4 \quad (5)$$

$$x = -\pi/6 + \pi/12 = -\pi/12 = -5\pi/12 \quad (5)$$

(45)